KERAUNOS

Electromagnetic Launch from the Moon, Mars, and Earth

Maglev Space Launch System · SELENE · BRONTE · ARES

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Perpetual Technologies · Engineering Whitepaper

Revision 2 — June 12, 2026. Fact-checked edition: every external claim re-verified against sources as of this date; physics re-derived; BRONTE site engineering replaced with the Chimborazo Design Map study (Annex A).

What Changed in Revision 2

Revision 1 (June 2026) was reviewed line-by-line. The physics held: every track length, velocity, energy, and acceleration figure re-derived cleanly. The empirical claims needed surgery. Material corrections, so readers of Rev 1 know what moved:

Corrected facts. SpaceX's lunar mass-driver remarks were made in **February 2026** (all-hands, Feb 11; reaffirmed Mar 22 on X) — they are Musk statements, not a published program plan · China's Ziyang EM platform targets "**above Mach 1**" before ignition, not Mach 1.6 · Auriga Space's **$12.2M is cumulative** funding (VC + AFWERX), not one round · X-59 flew its **first supersonic flight June 5, 2026** (Mach 1.1) — Rev 1 predated this · the all-REBCO magnet record is now **28.2 T** (CHMFL 2025) and the all-superconducting record **35.1 T** (ASIPP, Sept 2025) · NIF's best shot is now **8.6 MJ** (April 2025), not 3.15 MJ · the plasma window has been **demonstrated** for non-vacuum e-beam welding, not deployed industrially · the "600 million tonnes of ice in Shackleton" figure was a misattribution — it belongs to **north-pole** craters (Mini-SAR, 2010); Shackleton-specific surveys support only bounded estimates (≤5–10 wt% in wall regolith), with Chang'e-7 due to ground-truth the south pole from late 2026 · Falcon 9's 2026 list price is **~$74M (~$3,245/kg expendable)** · a GPS III launch ran **$83–97M**, not $60M · ISS spends **~$3.1B/yr** (~$1.8B on transport) · helium-3 trades near **~$20B/tonne**, and D-He3 is **low-neutron**, not strictly aneutronic · StarTram Gen-1's published exit is a **3–7 km** mountain peak · Chimborazo's exit-level air is **~52% of sea-level density** (45% is the pressure ratio) · phase numbering is now consistent throughout (SELENE 1 · Chimborazo 2 · Vacuum fleet 3 · ARES 4 · Full Earth injector 5) · the 80 t/day orbital cadence belongs to the **full Earth injector**, not the Chimborazo demonstrator.

Engineering replaced, not just corrected. The Chimborazo section is rebuilt on a terrain-and-geology study: real SRTM 30 m elevation data, an automated route fit, IG-EPN volcanic status, Pallatanga-fault seismicity (~0.4 g design basis), and TBM precedent. The result is a different and better machine than Rev 1 sketched: a single straight-then-bent line through the mountain with a crest exit at 6,160 m. Full drawings: **Annex A — BRONTE Chimborazo Design Map (KER-BRO-TRK-001 REV N)**, bound into this document after Section 4.

ALL DESIGN CONTENT REMAINS EARLY-VERSION CONCEPT MOCK-UP — ILLUSTRATES THE IDEA AND GOAL ONLY — NOT FOR CONSTRUCTION

0. Executive Summary

Earth, with its thick atmosphere, is the hardest body in the system to launch from electromagnetically. So the program starts on the Moon, where the machine is native, and works outward. The full Earth injector is the long-horizon prize. It is not the opening move.

Launcher	Body	Phase	Character
SELENE	Moon	1 — Proof of concept	No atmosphere: no tube, no plasma window, no aero-shell. Starter track 1–3 km; at 100 g, 2.9 km clears lunar escape in 2.4 s. The first segment is a working exporter, not a demo.
BRONTE	Earth	2 — The Chimborazo Project	A 14 km first build on a ~38 km through-mountain line at Chimborazo, Ecuador; crest muzzle at 6,160 m. Proves tube, plasma window, and aero-shell at sub-scale. DoD hypersonic test contracts pay for it.
Fleet	Airless bodies	3 — Vacuum-world kits	SELENE-class launchers stamped onto Phobos, Deimos, Ceres, the Belt. The core bet.
ARES	Mars	4 — The port	Mars atmosphere is 0.6% of Earth's; the tube is featherweight. ARES closes the return-trip problem every settlement plan dies on.
BRONTE full	Earth	5 — The endgame	The ~38 km Mach-12 line, then beyond — built only after Chimborazo Phase 1 validates cost and subsystems.

SELENE: Moon (Phase 1)

Parameter	Value	Notes
Track (PoC)	1–3 km, 50–100 g	At 100 g: 1.5 km = orbit speed (1.72 km/s); 2.9 km = escape (2.38 km/s) in 2.4 s
Track (cargo)	10–12 km, 30 g	2.43–2.66 km/s; escape with margin; ~9 s ride
Track (passengers)	~50 km, 3 g	1.72 km/s to low lunar orbit in 58 s (escape-rated 3 g line: ~96 km, future)
Energy per kg	0.41 kWh orbit / 0.78 kWh escape	Orbit ≈ 1/22 of Earth's 8.9 kWh/kg-to-LEO; escape ≈ 1/11
Atmosphere	None	No tube, no plasma window, no aero-shell. The track is the launcher.
Cooling	20 K closed-loop cryoplant	Sealed cryostats hold every coil at 20 K; shadowed cold sinks and ~250 K regolith at 2 m depth help — sunlit surfaces swing far hotter, so the cryostat does the real work

BRONTE: Earth (Phase 2 — Chimborazo Project)

Parameter	Value	Notes
Site	Chimborazo, Ecuador	Summit 6,263 m; 1.47°S; massif sits on a 4,000 m plateau
Phase-1 build	14 km, portal 4,069 m	Eastern end of the full line; the only at-grade start the terrain allows

Exit	Crest muzzle 6,160 m, 12.3° east	Terminal bend R = 100 km over final 11 km; ~100 m below peak elevation
Exit air	$\rho \approx 53\%$ SL ($p \approx 46\%$)	Peak drag and heating roughly halved; weaker ground-level boom
Equatorial bonus	+465 m/s free	Fires east; net launcher requirement to LEO ~7.5 km/s, not ~8
Performance	Mach 8.9 (30 g) / 2.8 (3 g)	Full 38 km line later: Mach 12.6 cargo / 4.6 at human-rated 3 g
Revenue	$50K–$500K per test run	Hypersonic test services from day one — the market Auriga Space is already selling into

ARES: Mars (Phase 4)

Parameter	Value	Notes
Atmosphere	~6 mbar (0.6% of Earth)	Featherweight tube; the planet does 99.4% of the plasma window's job
Track (orbit)	~180 km, 3.3 g	Low Mars orbit (~3.4 km/s) in a 105 s run
Track (escape)	~390 km, 3.3 g	Mars escape (~5.0 km/s) in a 155 s run
Energy per kg	1.6 kWh orbit / 3.5 kWh escape	Between Moon and Earth in scale; far easier than Earth in engineering
Staging	Phobos and Deimos	Catch points and depots; SEP tugs fly the interplanetary legs

1. SELENE: The Moon Machine

The Moon has no atmosphere. That sentence deletes every hard engineering problem on the Earth list.

On the Moon these things do not exist: the vacuum tube (the track is already in vacuum), the plasma window (no atmosphere to keep out), the aero-shell, the blast shutters, the transpiration cooling. The track lays on regolith, braced and aligned. That is the system.

The starter track is **1–3 km** — airport-runway length. At 100 g, a 1.5 km run reaches lunar orbit speed (1.72 km/s); a 2.9 km run clears escape (2.38 km/s) in 2.4 seconds. Water, regolith, and metal don't mind the g-load. The very first segment isn't a demo. It is a working exporter.

> **0.78 kWh per kg to escape.** Lunar escape costs roughly one-eleventh of Earth-to-LEO launch energy (0.78 vs 8.9 kWh/kg at 8 km/s); lunar orbit, about one-twenty-second. From the Moon you are already partway to everywhere: a Mars-transfer trajectory from the lunar surface costs ~3.0 km/s total — about **1.25 kWh/kg**, roughly one-seventh of Earth-to-LEO energy, and roughly one-fifteenth the energy of flying the same trip up from Earth's surface.

> **The south pole is the right site.** Shackleton-rim and Connecting-Ridge sites see sun 80–90% of the time; permanently shadowed cold traps sit next door; Earth hangs permanently on the horizon for communications. Rev 2 honesty on two Rev 1 claims: polar *surface* temperatures swing widely (shadow cryo-cold to ~200 K+ in sun), so the 20 K coils live in sealed cryostats regardless — the real thermal gift is vacuum, cold sky, and ~250 K regolith two meters down. And the ice: the famous "600 million tonnes" belongs to *north-pole* crater radar (Mini-SAR 2010). For Shackleton itself the honest statement is bounded — radar and reflectance permit up to ~5–10 wt% ice in wall regolith, with no credible tonnage yet; Chang'e-7 (launch ~Aug 2026) goes to ground-truth exactly this. The architecture survives the uncertainty: regolith, oxygen, and metal exports close the early business even in the lean-ice case.

The physics has been settled for fifty years. Gerard O'Neill's NASA Ames summer studies (1976–77) developed the lunar mass driver; Mass Driver One was demonstrated in 1977. A San José State University engineering study (E. Miller, 2023) sizes a polar machine launching 25.4 kg pellets at 2.4 km/s every 10 seconds on 8.7 MW of solar power, ~1.0 MA per bucket coil — SELENE's starter track is exactly that class of machine. What killed every previous program was the same thing: all government funding, no revenue until a Moon base already existed. Cislunar freight removes that deadlock.

The product is propellant — and mass itself. Polar ice (wherever the surveys land), electrolyzed to LOX/LH2 or processed with imported carbon to methalox, becomes propellant for every vehicle in cislunar space, delivered to L1/L2 depots at roughly 10× less than lifting the same kilogram from Earth. Regolith is radiation shielding; sintered metal is structure. The Moon doesn't need factories to matter — it needs to be a source of cheap mass, and it already is.

SELENE grows by ladder — 1–3 km PoC → 10–12 km / 30 g hardened cargo → ~50 km / 3 g passenger line — each step funded by the previous segment's exports. The SELENE kit is the template for every airless body worth working: Phobos, Deimos, Ceres, the Belt. Design once, stamp copies. The fleet is the real bet.
(Companion sheet: SELENE Design Map, KER-SEL-TRK-002 — separate document.)

2. The Physics and the Price

The electricity cost of orbital velocity is $0.53 per kilogram. Everything above that is hardware amortization, not physics.

Relation	What it means
$E = ½v^2$ = 32 MJ/kg	At 8 km/s and $0.06/kWh: **$0.53 of grid electricity per kilogram**. Falcon 9 burns ~$200–300K of propellant for ~17–22.8 t to LEO — propellant alone costs ~20–25× KERAUNOS's total electricity spend per kilogram, on a vehicle that is 90% propellant by mass.
$a = v^2/2L$	Acceleration is set by velocity and length. 1,000 km at 8 km/s is 3.3 g — gentle enough for people. StarTram Gen-1 accepted 30 g on 130 km, cargo-only. On the Moon a 3 g orbital track is ~50 km. **Track length is the unlock.**
Kick stage: Isp 350 s, 20% mass	Δv = 3,434 × ln(1/0.8) ≈ **0.77 km/s** for circularization and trim at apogee. That is the only propellant aboard — ~80% of launched mass is payload, vs ~10% for a rocket.
Bends are forbidden at speed	Curving 20° at 4 km/s needs a ~200 km radius. Every high-speed launcher is a straight line with, at most, a gentle payload-g-limited terminal bend. This single rule shaped the entire Chimborazo route (Section 4).

Comparative launch costs (per kg to LEO)

System	Cost per kg	Economic bottleneck
Space Shuttle	$54,500 (full-program-cost basis)	Hand-built; massive refurbishment every flight
Falcon 9 (2026)	~$3,245 list (expendable basis); ~$630 est. internal	$74M list price (Feb 2026); reusable flights carry ~17–18 t, so real per-kg runs higher
Starship (projected)	$100–$1,000 — **aspirational**	V3 flew once (Flight 12, May 22, 2026; payload deploy succeeded, booster mishap, fleet grounded pending FAA review). No reuse-economics data yet; chemical energy density is the hard floor
KERAUNOS	$20–$250 (target)	$0.53/kg electricity; the rest amortizes track capital — the honest unknown is tunnel+tube cost per km, which Phase 1 exists to measure
KERAUNOS + skyhook	toward $5	Kick propellant → ~2–5% when a rotating tether catches pods (Section 8)

3. Why Now: The Signals, as of June 2026

The field validated itself publicly in the last 24 months. The window to enter at peer level is now.

- **SpaceX wants a lunar mass driver — said out loud, February 2026.** Following the company's lunar pivot (Feb 2, 2026), Musk described a mass driver "shooting AI satellites into deep space" at the Feb 11 all-hands, and reaffirmed it March 22 ("Mass drivers on the Moon!"). Honesty requires the caveat: these are founder statements, not a funded program of record — no contracts or milestones have been disclosed. But the largest launch company on Earth pointing at exactly this machine is both validation and a clock. A partner positioned ahead of that build doesn't compete — it supplies the architecture.
- **China is building EM launch hardware, targeting 2028.** Galactic Energy, CASIC, and the Ziyang municipal government are constructing an electromagnetic launch verification platform to throw a Ceres-2-class rocket to **above Mach 1** before ignition. Progress is real: a full-system ejection test of a 1.4 m rocket model (April 2025) and a matrix-switched segmented-drive trial at ~90% system efficiency (January 2026). It's launch-assist — the rocket still carries the full propellant penalty — but the supply chain and urgency are real.
- **The US military already funds this class of technology.** Auriga Space has raised $12.2M cumulatively (VC plus AFWERX, including a $1.25M Direct-to-Phase-II SBIR) for an EM launch-assist track, selling hypersonic test services as Phase-1 revenue — Prometheus (lab) → Thor (outdoor track, 2026) → Zeus (orbital). General Atomics EMALS (484 MJ flywheel storage, 45 s recharge) has launched aircraft from USS Gerald R. Ford since 2017 through a combat deployment. A 2023 AFOSR-filed report (Peterkin) explicitly recommends evolving EMALS for lunar launch, and a US SBIR program record describes a magnetically levitated sled driven to 7 km/s. The demand signal is public and funded.
- **The boom-shaping toolkit just went supersonic.** NASA's X-59 flew Oct 28, 2025 (67 min), expanded its envelope through spring 2026, and made its **first supersonic flight June 5, 2026** (Mach 1.1), with Mach 1.4 acoustic-validation points next — flying on a camera-vision cockpit the whole way. The quiet-shock shaping ASTRAPE borrows is being flight-proven right now.
- **The gap nobody fills:** every competitor is fighting the atmosphere — China's launch-assist, StarTram's paper tube, Auriga's track. Nobody is building the launcher where the atmosphere doesn't exist. KERAUNOS starts there; the vacuum-world fleet is the open lane, and Earth revenue (Chimborazo) funds the climb.

4. The Chimborazo Project: BRONTE's Earth Demonstrator

Rev 2 replaces the sketch with a site study: real terrain, real geology, one straight line through the mountain.

Chimborazo, Ecuador. Summit 6,263 m, 1.47°S. The case still holds: the equator gives +465 m/s of Earth's rotation free (fire east); high exit altitude halves the air (at 6,160 m, density is ~53% of sea level — Rev 1 quoted 45%, which is the *pressure* ratio); equatorial launch reaches every inclination. What changed is the engineering, driven by three facts established in the Rev 2 site study:

- **Terrain (SRTM 30 m):** the mountain is a gentle 40-km shield with a concave western approach — no straight surface route to a summit-area exit exists. The answer is a **through-mountain line**: azimuth 095°, passing 0.5 km north of the summit.
- **Geology (IG-EPN, Bernard et al., Beauval et al.):** the andesite core is TBM-proven rock (Ecuador's own Coca Codo Sinclair bored 24.8 km with 9 m machines); the volcano is dormant-but-potentially-active (last eruption ~550 CE, ~1,000-yr recurrence); the governing hazard is seismic — the Pallatanga fault (1797 Riobamba, M~7.6) sets a ~0.4 g design basis, which drove A-frame piers, radial truss bracing, and re-trimmable track jacks. The route avoids the Riobamba debris-avalanche deposit entirely.
- **The straight-line rule:** at exit speeds the track cannot bend more than a payload-g-limited R = 100 km terminal arc — which buys the final 11 km a climb from 6.5° to a 12.3° exit and lifts the muzzle ~550 m to the ridge crest.

The line, in two builds

	Phase 1 — build first	Full line — the growth target
Length	**14 km** (portal 4,069 m — the only at-grade start the terrain allows)	**~38 km** (west portal ~1,360 m, Chazojuán valleys)
Works	El Arenal viaduct (A-frame piers 110–590 m) + 1.2 km crest bore + truss muzzle	adds 24 km of deep tunnel, cover ≤ ~880 m (Gotthard runs 2,300 m)
Exit	Same crest muzzle for both: **6,160 m**, 12.3° east, ~100 m below peak elevation	
Cargo (30 g)	2.81 km/s · **Mach ~8.9** · 10 s in tube	3.97 km/s · **Mach ~12.6** · 16 s (16 g bend cap)
Human (3 g)	Mach ~2.8 · 31 s	Mach ~4.6 · 50 s (2.2 g in the bend)
Per shot (15 t gross)	16.5 MWh · 12.4 GW peak	32.8 MWh · 17.5 GW peak — trackside storage; grid sees a 30 MW × ~66 min recharge

The tube itself: Ø4.0 m flight bore inside a Ø6.0 m ring-stiffened vacuum vessel inside a Ø12 m TBM bore, driven by a **360° stator ring** (24 REBCO sectors per ring, hoops every 0.6 m, ~63,000 on the full line) that levitates, centers, steers, and propels from all sides — required, not optional, because the bend presses the 15 t vehicle outward at up to 16 g. The payload bay is **Ø3.2 m × 8 m** (Cygnus-module class) inside the 23 m ASTRAPE fairing-sled. Vacuum isolation gates every 500 m; 20 K/80 K cryo headers full-length with cooler stations every 500 m.

The honest open issues, on the record: El Arenal piers reach ~590 m (Millau's record is 343 m); the plasma window is proven only at centimeter scale and the Ø4 m bore costs 2.8× the baseline window power — it gets the first R&D dollar; the whole massif sits inside the Reserva de Producción de Fauna Chimborazo under Ecuador's rights-of-nature constitution — the legal path may be the hardest engineering on the sheet; and Mach-12 exits are loud regardless of shaping — altitude and the empty eastern corridor are the real mitigations, and test recovery overflies the inter-Andean valley like any sounding-rocket range.

Revenue from day one

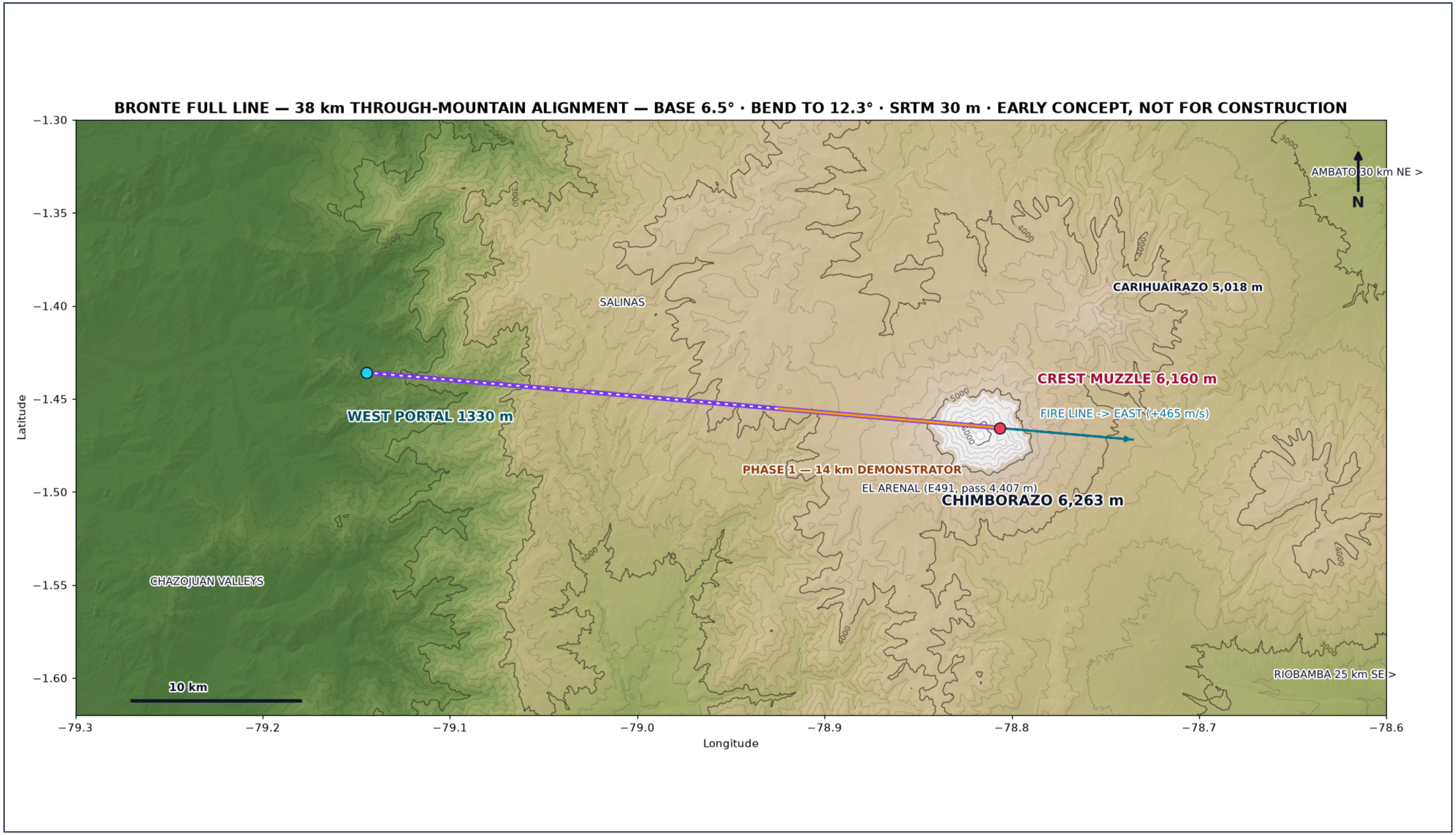
DoD hypersonic test services: sounding rockets are single-shot, wind tunnels don't hold real Mach 6+ conditions for minutes, and no facility on Earth fires Mach-9-class test articles from 6,160 m on the equator several times a day. At $50K–$500K per run, one run per week covers operations. The demonstrator is not a cost center — it is a product, and every shot also retires risk on the plasma window, the aero-shell, and the cost-per-kilometer number the full injector's economics live or die on.

ANNEX A follows this page: the complete BRONTE Chimborazo Design Map (KER-BRO-TRK-001 REV N) — route map on real terrain, both build profiles at true 1:1 scale, tunnel & tube sections, the 360° drive ring, ASTRAPE in three views, and nine engineering charts. Every figure on it re-derived and audited June 12, 2026.

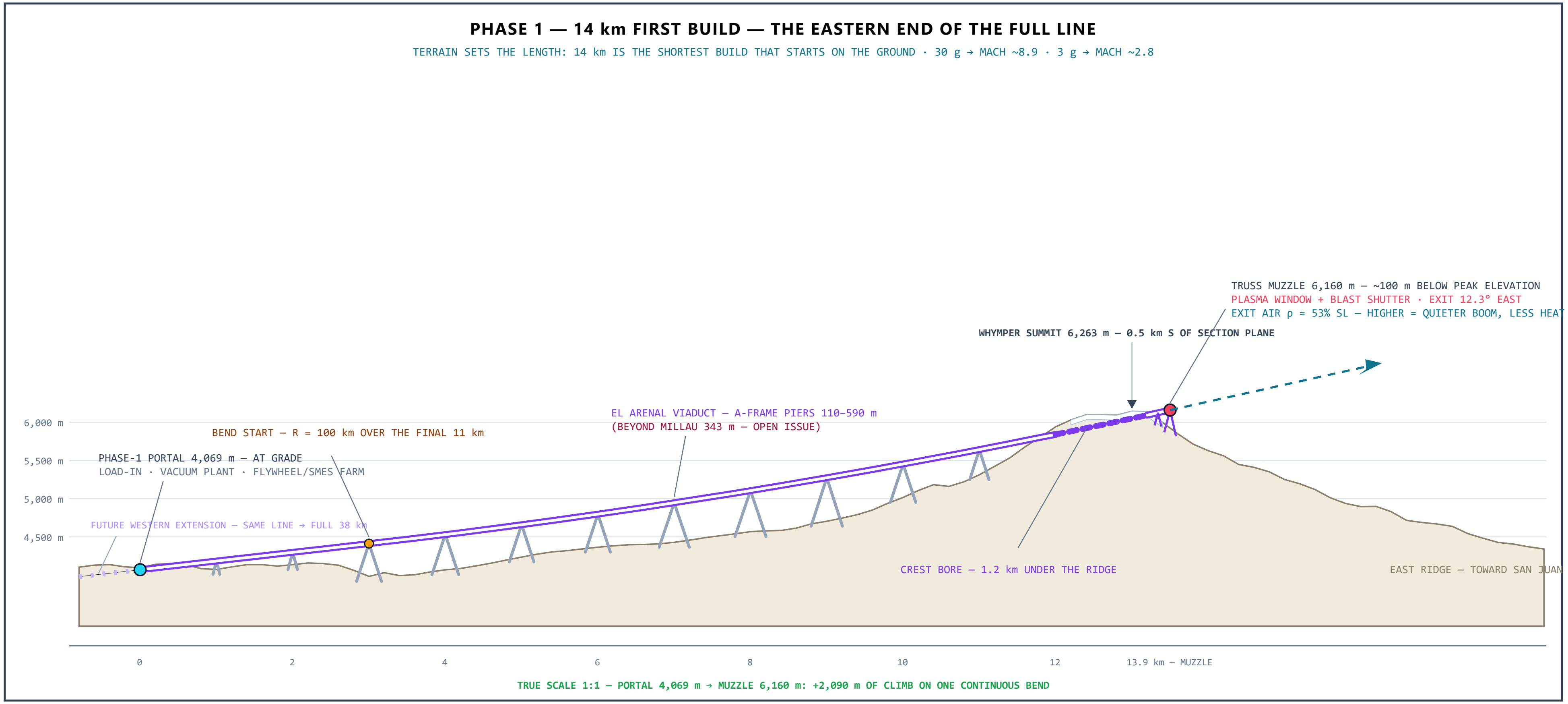
⚠ EARLY-VERSION TEST MOCK-UP — ILLUSTRATES THE IDEA AND THE GOAL ONLY · NOT A DESIGN · NOT FOR CONSTRUCTION · EVERYTHING SUBJECT TO COMPLETE CHANGE ⚠

VERY EARLY CONCEPT ILLUSTRATION — NOT A DESIGN DOCUMENT. The complete BRONTE Chimborazo design map in one sheet: route (real SRTM 30 m terrain), both build phases, the tunnel & tube section with the 360° drive ring, and engineering charts for both. Tube architecture: full-circumference stator — levitation, centering, and propulsion from all sides — sized for a Ø3.2 m payload. All dimensions speculative and unvalidated.

PANEL A — PLAN VIEW: ONE LINE, TWO PHASES (SRTM 30 m)



PANEL B — PHASE 1 (BUILD FIRST): 14 km — PORTAL 4,069 m → CREST MUZZLE 6,160 m (TRUE 1:1)



PANEL C — FULL LINE (THE GROWTH TARGET): ~38 km — 27 km STRAIGHT @ 6.5° + 11 km BEND TO 12.3° (TRUE 1:1)

FULL LINE — WEST VALLEYS → DEEP TUNNEL → VIADUCT IN THE BEND → CREST MUZZLE 6,160 m

WEST PORTAL ~1,360 m (CHAZOJUÁN) · ROCK PORTAL ~6,090 m · TRUSS MUZZLE 6,160 m (PEAK 6,263 m) · EXIT 12.3° EAST · TRUE SCALE 1:1

GENERAL NOTES

1. HIGH-SPEED TRACK CANNOT BEND HARD. R = 100 km OVER THE FINAL 11 km IS THE PAYLOAD-g LIMIT, NOT A COST CHOICE. EXIT ANGLE 6.5° → 12.3°.
2. WHY ALTITUDE: EXIT AIR @ 6,160 m IS ρ ≈ 53% SL (p ≈ 46%). EVERY METER HIGHER MEANS A WEAKER GROUND-LEVEL BOOM, LESS DRAG, LESS HEATING.
3. PRICE OF THE CREST EXIT: EL ARENAL PIERS REACH ~590 m (MILLAU RECORD 343 m). A-FRAME PIERS FOR SEISMIC STIFFNESS — OPEN ISSUE.
4. WEST DEEP TUNNEL 24 km, COVER ≤ ~880 m (GOTTHARD: 2,300 m).
5. SEISMIC ~0.4 g (PALLATANGA) · VOLCANO DORMANT (~550 CE) · RESERVE LEGAL STATUS OPEN · FIRES EAST: +465 m/s ROTATION BONUS FREE.
6. REV H CORRECTION: 6,255 m MUZZLE RETRACTED (NEEDED ~520 m MAST).

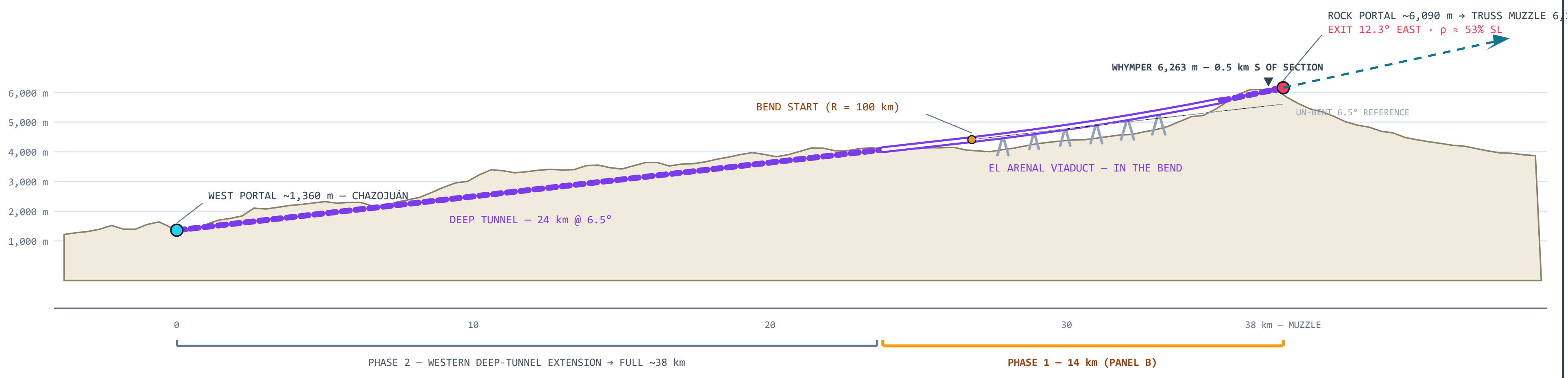
PERFORMANCE — BASELINE 15 t · MAX GROSS 20 t

FULL — 30 g + BEND COAST → 3.97 km/s · MACH ~12.6
FULL — 3 g HUMAN (DRIVE-THROUGH) → 1.45 km/s · MACH ~4.6
PHASE 1 — 30 g → 2.81 km/s · MACH ~8.9
PHASE 1 — 3 g → 0.89 km/s · MACH ~2.8

BEND NORMAL g: 16 CARGO CAP · 8 PHASE-1 · 2.2 HUMAN
BEND TRADES ~2 MACH (CARGO) FOR +550 m EXIT ALTITUDE

TIME IN THE TUBE (FROM REST TO EXIT):

30 g — 16 s FULL · 10 s P1 5 g — 39 s FULL (M6.0) · 24 s P1 (M3.6)
3 g — 50 s FULL (M4.6) · 31 s P1 (M2.8)
AT 20 t MAX: THRUST 5.9 MN · 43.7 MWh · PEAK 23.4 GW

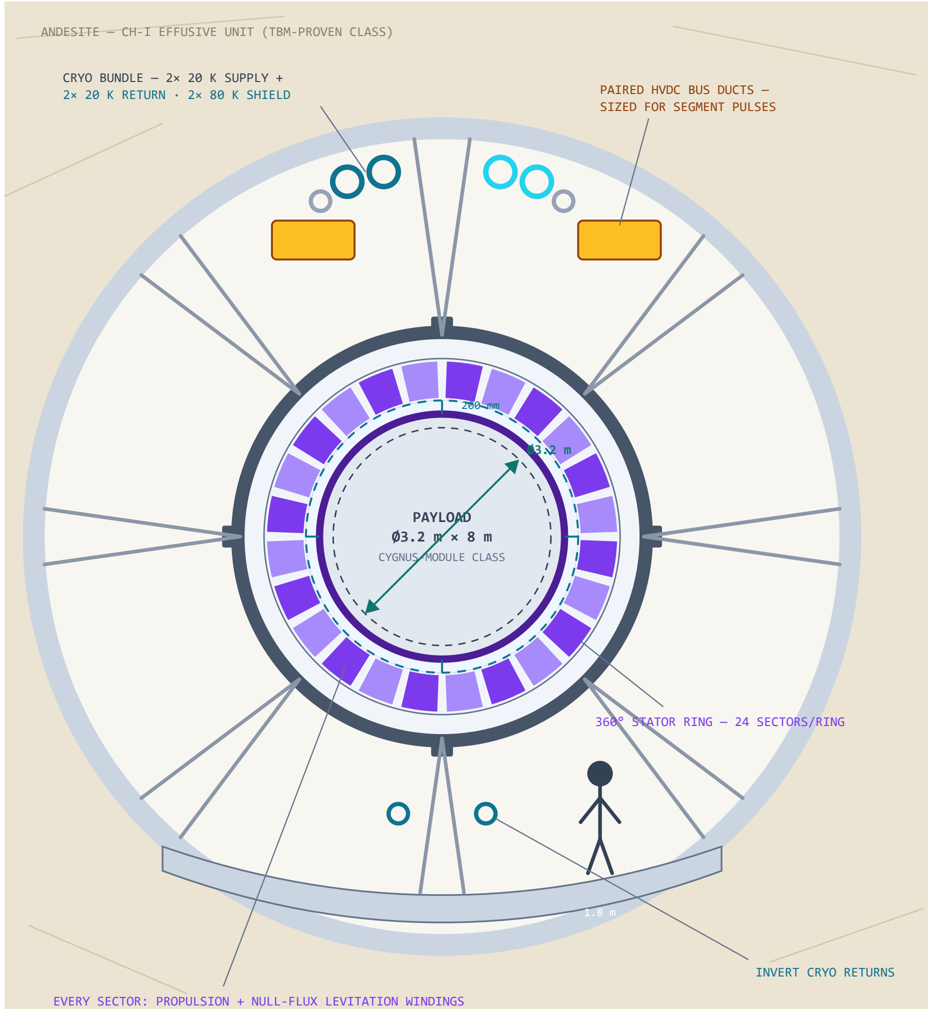


TRUE SCALE 1:1 — WATCH THE TRACK PEEL UP FROM THE DASHED 6.5° REFERENCE LINE THROUGH THE FINAL 11 km

PANEL D — TUNNEL & TUBE TYPICAL SECTION — 360° DRIVE RING, WIDE BORE FOR Ø3.2 m PAYLOAD

INSIDE THE TUNNEL — FULL-CIRCUMFERENCE STATOR: LEVITATE, CENTER, AND PROPEL FROM ALL SIDES

Ø12.0 m TBM BORE · Ø6.0 m VACUUM VESSEL · 360° STATOR RING Ø4.0–5.2 m · Ø4.0 m FLIGHT BORE · Ø3.6 m VEHICLE · GAP 200 mm ALL ROUND · DRAWN TO SCALE



EVERY SECTOR: PROPULSION + NULL-FLUX LEVITATION WINDINGS

RADIAL TRUSS SPOKES — VESSEL BRACED 360° TO THE LINER AT EVERY RING STATION (SEISMIC + ALIGNMENT)

WHY THE RING MUST BE 360° — NOT A FLOOR GUIDEWAY

IN THE 11 km BEND THE VEHICLE PRESSES OUTWARD AT UP TO 16 g — 2.35 MN ON THE 15 t VEHICLE. GRAVITY (1 g) IS ROUNDING ERROR. THE "LOAD WALL" IS WHEREVER THE CURVE POINTS, NOT THE FLOOR. A FULL-CIRCUMFERENCE STATOR CARRIES THRUST, LEVITATION, AND CENTERING IN EVERY DIRECTION AT ONCE — AND DOUBLES AS THE STEERING THAT FLIES THE VEHICLE THROUGH THE BEND ON RAILS OF FIELD INSTEAD OF METAL.

SECTION STACK — SIZED FROM THE PAYLOAD OUT

Ø12.0 m TBM BORE — PRECEDENT: BERTHA 17.45 m · CCS 9.04 m

0.32 m SEGMENTAL LINER + GROUT

Ø6.0 m VACUUM VESSEL — 40 mm 316L + RINGS @ 2.5 m
SIZED BY EXTERNAL CRUSH (BUCKLING), NOT BURST

Ø4.0–5.2 m STATOR RING — 24 SECTORS × HOOPS EVERY 0.6 m
~63,000 COIL HOOPS ON THE FULL 38 km LINE

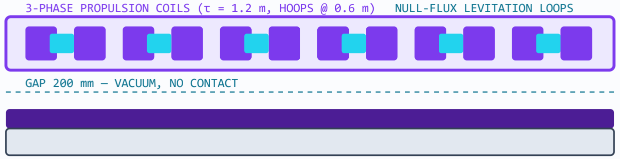
Ø4.0 m FLIGHT BORE · ~10 Pa · GAP 200 mm ALL ROUND

Ø3.6 m VEHICLE “ASTRAPE-H” — 23 m · 15 t (MAX GROSS 20 t) · FULL-HULL SLEEVE

Ø3.2 m × 8 m PAYLOAD BAY — CYGNUS-CLASS FITS WITH MARGIN

PRICE OF WIDE: PLASMA WINDOW POWER SCALES WITH AREA — Ø4.0 m ≈ 2.8× THE Ø2.4 m BASELINE. CHEAP IN ROCK, DEAR AT MUZZLE.

RING SECTOR ZOOM — ONE OF 24, UNROLLED FLAT

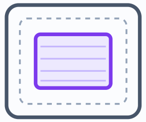


VEHICLE REACTION SLEEVE (HALBACH + CONDUCTOR) — FULL HULL

PASSIVE FAILSAFE: NO DRIVE → VEHICLE SETTLES ONTO SKID RAILS AT BORE BOTTOM

EACH SECTOR: OWN CRYOSTAT BRANCH · OWN QUENCH DUMP · INDIVIDUALLY SWITCHED

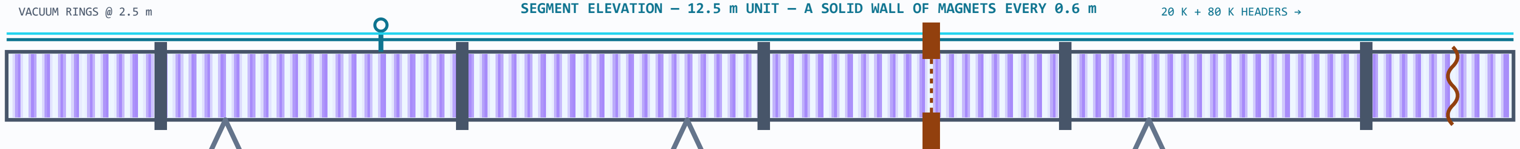
STATOR MAGNET MODULE — ONE COIL



VACUUM CASE 316LN · MLI + 80 K SHIELD
NI REBCO DOUBLE-PANCAKES @ 20 K
VPI EPOXY · 316LN FORMER · SPARC-CLASS BUILD
ONE LOOP · ONE CRYO BRANCH · ONE QUENCH DUMP

FIELD CLASS 5–10 T AT THE GAP — A LAUNCH LSM NEEDS FORCE, NOT FUSION-GRADE FIELD

SEGMENT ELEVATION — 12.5 m UNIT — A SOLID WALL OF MAGNETS EVERY 0.6 m



DRIVE HOOPS @ 0.6 m — ~21 PER SEGMENT, ~63,000 ON THE FULL LINE · ALIGNMENT JACKS EVERY ~5 m · ISOLATION GATE @ 500 m — BREACH VENTS 500 m, NOT 38 km · BELLOWS

CRYO FEED STUB (TOP) — COOLER STATION EVERY 500 m · 20 K + 80 K HEADERS RUN THE FULL LENGTH

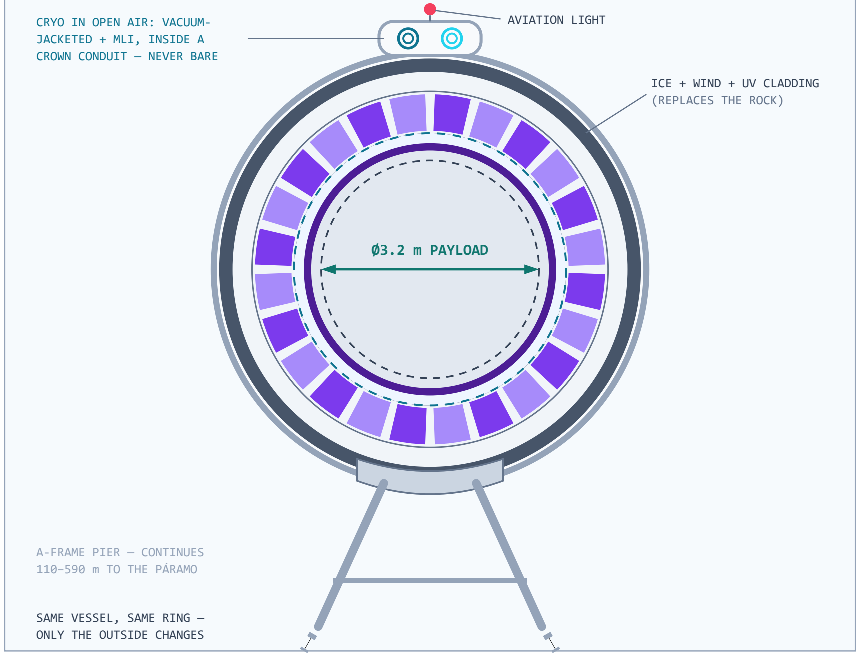
PANEL D CONCEPT ONLY — KER-BRO-TUN-001 FOLDED INTO THIS SHEET — EVERY DIMENSION SPECULATIVE

PANEL E — EXTERNAL (BRIDGE) SECTION · ASTRAPE FRONT VIEW · ASTRAPE IN THE TUBE, SIDE PROFILE

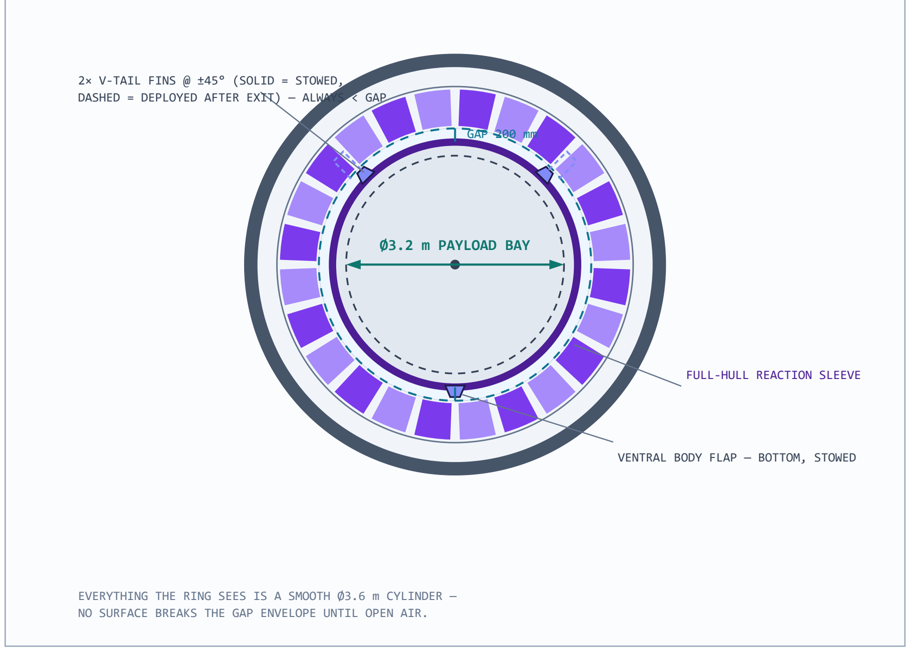
THE SAME TUBE, OUTSIDE THE MOUNTAIN — AND THE VEHICLE SHOWN IN IT, FRONT AND SIDE

LEFT: VIADUCT ("BRIDGE") SECTION ON A-FRAME PIER · RIGHT: ASTRAPE FRONT VIEW IN THE BORE · BOTTOM: SIDE PROFILE WITH LENGTH DIMENSIONS

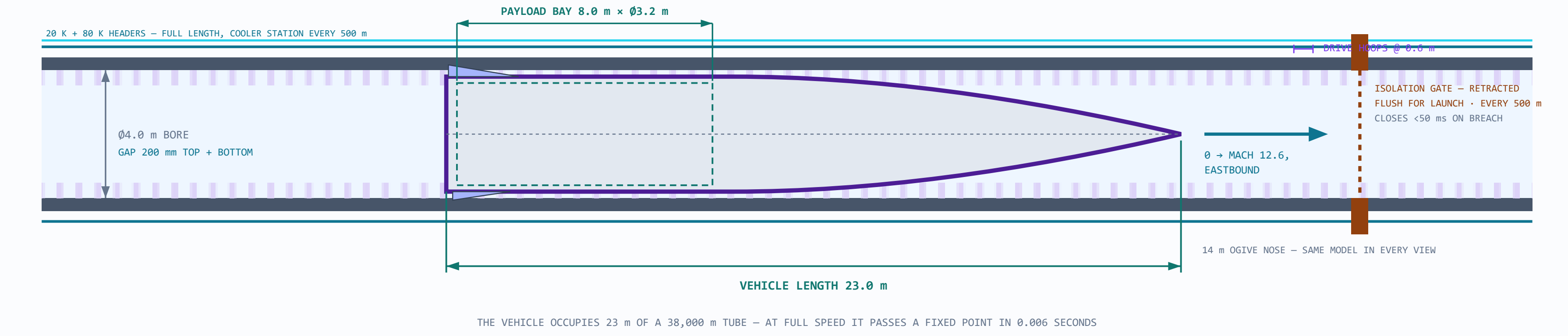
SECTION ON THE EL ARENAL VIADUCT — OPEN AIR



ASTRAPE — FRONT VIEW, INSIDE THE BORE (IN TUNNEL)

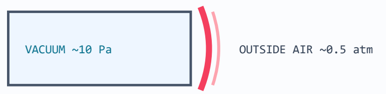


ASTRAPE IN THE TUBE — SIDE PROFILE, CUTAWAY (SCALE 30 px/m · 46 m OF TUBE SHOWN)



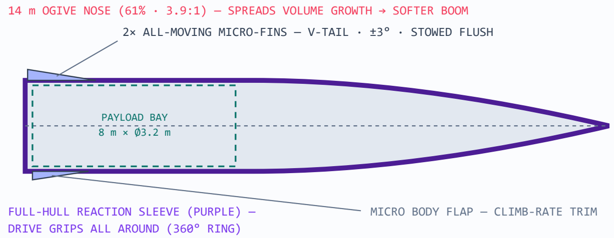
DETAILS — SYSTEMS

WHAT THE PLASMA WINDOW ACTUALLY IS



A SHEET OF IONIZED GAS (~12,000–15,000 K PLASMA) HELD ACROSS THE MUZZLE BY MAGNETIC FIELDS. ITS PRESSURE AND VISCOSITY BLOCK AIR FROM RUSHING INTO THE VACUUM TUBE — BUT A SOLID OBJECT FLIES STRAIGHT THROUGH IT. AN INVISIBLE DOOR THAT IS NEVER IN THE WAY, SO THE TUBE STAYS EVACUATED TO THE INSTANT OF EXIT. INVENTED BY A. HERSHCOVITCH, BROOKHAVEN (PAT. 1995); HOLDS ~9 atm AT cm SCALE. SCALING TO THE Ø4.0 m WIDE BORE IS UNPROVEN — FIRST R&D DOLLAR; PHASE 1 TESTS IT AT FULL BORE.

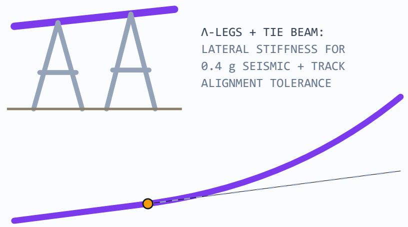
FAIRING-SLED “ASTRAPE” — THE FAIRING IS THE SLED



AT MACH 9–13, FINGERNAIL-SIZED DEFLECTIONS STEER HARD — SMALL FINS ARE THE DESIGN.

ASTRAPĒ — LIGHTNING, SISTER OF BRONTE · MACH 12 FINENESS — SAME MODEL AS PANEL E

A-FRAME PIER + TERMINAL BEND ×10

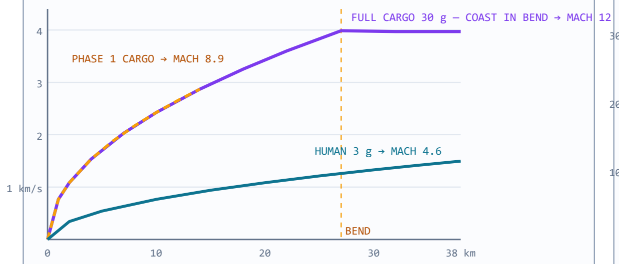


R = 100 km · FINAL 11 km · 6.5° → 12.3° · LIFT ~550 m

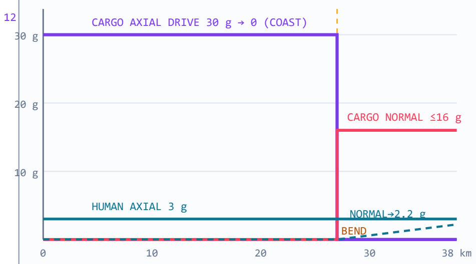
NORMAL g: 16 CARGO (COAST) · 8 PHASE-1 · 2.2 HUMAN

DETAILS — ROUTE ENGINEERING CHARTS

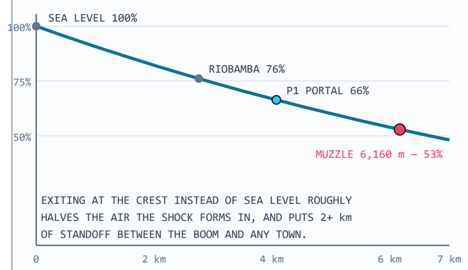
VELOCITY vs TRACK DISTANCE



g-LOAD PROFILE vs TRACK DISTANCE (FULL LINE)

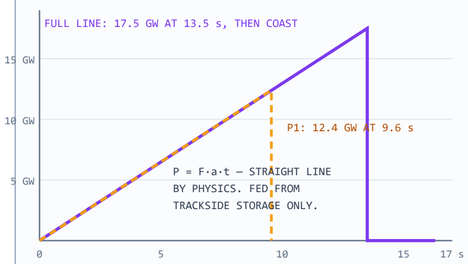


AIR DENSITY vs ALTITUDE — WHY EXIT HIGH

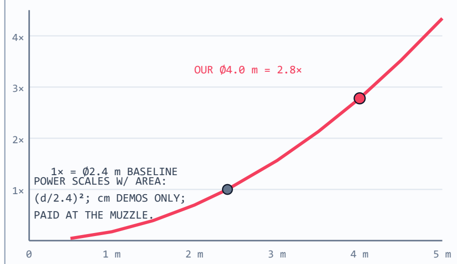


DETAILS — SYSTEMS CHARTS

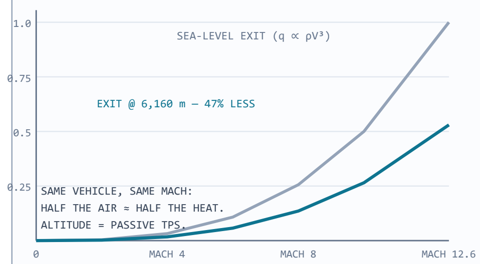
ONE SHOT, AS THE POWER SYSTEM SEES IT



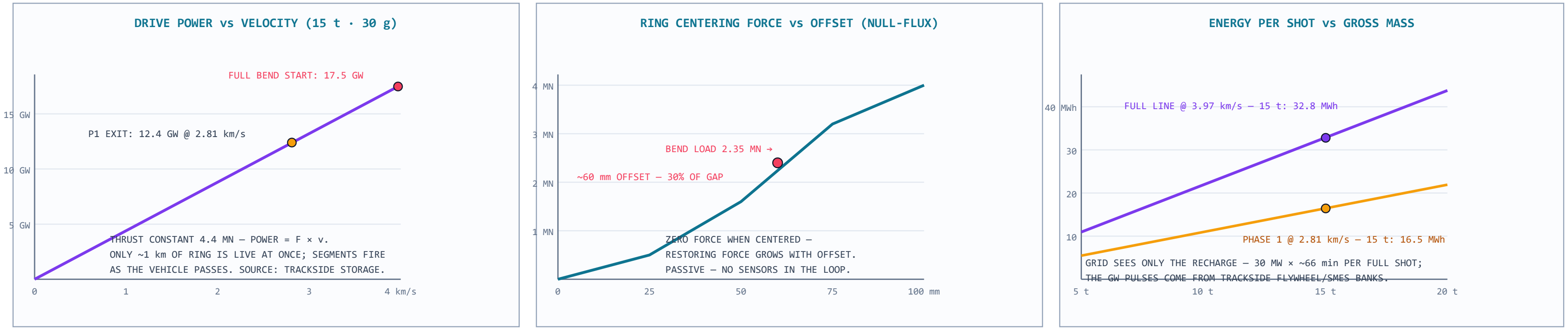
PLASMA WINDOW POWER vs BORE DIAMETER



NOSE HEAT FLUX vs MACH — WHY EXIT HIGH, AGAIN



DETAILS — TUNNEL ENGINEERING CHARTS



PERPETUAL TECHNOLOGIES — KERAUNOS	BRONTE — CHIMBORAZO DESIGN MAP — ROUTE + TUNNEL + VEHICLE	KER-BRO-TRK-001	REV N	1:1	2026-06-12	EARLY MOCK-UP — NOT FOR CONSTRUCTION

*REV N — catwalks deleted (the tube is sealed; service is robotic, so nothing for people to stand on), aviation-light label moved off the panel title, and a third charts row added for the support systems. Prior pass: max tonnage set (20 t gross ceiling, 15 t baseline), tube-transit times added for 3/5/30 g, 360° radial truss bracing between vessel and borehole liner, vacuum-jacketed cryo conduit shown for open-air bridge spans, overlaps cleaned, and the ogive softened a touch. Prior note: One vehicle model everywhere now: 23 m total — a 14 m tapering ogive needle nose (3.9:1 nose fineness, stretched specifically to spread the volume growth and soften the sonic boom; it cuts wave drag too) on a 9 m full-diameter cylinder (the math forces the cylinder: an 8 m × Ø3.2 m bay can only pack into the full-diameter aft section, which is exactly 9 m long). Vacuum isolation gates every 500 m sectionalize the tube — retracted flush for launch, closing in under 50 ms on a breach, so a puncture vents 500 m of tube instead of 38 km; one is shown retracted in the cutaway. Cryogenics grew to a real system: paired 20 K supply and return headers plus an 80 K shield loop running the full length, cooler stations every 500 m, and invert return lines in the section view. The HVDC bus ducts doubled up and sized up. Earlier revision notes follow. Panel E: the external ("bridge") section showing the identical vessel-and-ring stack riding an A-frame pier in open air with weather cladding instead of rock; ASTRAPE's front view in the bore with the V-tail fins stowed flush (they deploy only after exit — nothing breaks the 200 mm gap envelope in-bore); and the side-profile cutaway placing the full 18 m vehicle inside the tube with explicit dimensions — 18.0 m length, 8.0 m × Ø3.2 m payload bay, Ø4.0 m bore — against the 0.6 m drive-hoop pitch. The payload diameter now carries explicit dimension arrows in every section view. The tunnel section (formerly its own drawing) now lives as Panel D, rebuilt around your call: the drive is a **full 360° stator ring** — 24 sectors per ring, rings every 0.6 m, roughly 63,000 coil hoops on the full line — because the engineering agrees with you. In the 11 km bend the vehicle presses outward at up to 16 g (2.35 MN); gravity is rounding error, so the "load wall" is wherever the curve points and a floor guideway can't carry it. The ring levitates, centers, propels, and steers from every direction at once; the vehicle answers with a full-hull reaction sleeve (ASTRAPE's inset updated to match — no belly skid, nothing jettisoned). The bore tightened from Ø5.0 to Ø4.0 m for proper 200 mm magnetic coupling all round — your Ø3.2 m payload is untouched, and the plasma window bill actually improves (2.8× baseline instead of 4.3×). New tunnel charts at the bottom: drive power vs velocity (12–18 GW pulses, only ~1 km of ring live at a time), the null-flux centering curve showing the vehicle riding ~60 mm off-center through the bend with no sensors in the loop, and energy-per-shot vs mass with the grid only ever seeing a quiet sub-hour recharge.*

5. ARES: The Mars Port (Phase 4)

Mars atmosphere is 0.6% of Earth's. The hard parts of the Earth machine are nearly free there.

The tube holds back 0.6% of the pressure the Earth tube fights; the plasma window's job is 99.4% done by the planet; exit heating is a fraction of the Earth case. Dust storms, not aerodynamics, are the main environmental design driver.

Element	Figure	Notes
Orbit track	~3.4 km/s → ~180 km @ 3.3 g	105-second run to low Mars orbit
Escape track	~5.0 km/s → ~390 km @ 3.3 g	155-second run; between Moon and Earth in scale, far easier than Earth in engineering
Energy	1.6 / 3.5 kWh per kg	orbit / escape
Staging	Phobos & Deimos	Catch points and propellant depots — the warehouse district; SEP tugs fly the cruise

Every Mars mission arrives needing return propellant. ISRU solves it on the surface (Sabatier methalox from CO_2 and water ice); ARES solves it in orbit — propellant rides to Phobos at electromagnetic cost and waits in a depot for any departing vehicle. Return cargo, manufactured exports, and samples ride the same track. **ARES makes Mars a place you can leave as easily as you arrive.**

> **The Moon as deep-space launch platform.** A Mars-transfer from the lunar surface costs ~3.0 km/s (≈1.25 kWh/kg) — roughly one-fifteenth the energy of the same trip from Earth's surface. Anything Mars-bound that can be built on, or staged through, the Moon ships from the lunar track: relay satellites, sensor networks, pre-positioned supplies. Starship carries crew and cargo that must arrive fresh; the mass drivers send everything else ahead, early and cheap. The Mars supply chain runs through the Moon, not around it. ARES is Phase 4: it inherits a mature vacuum-world kit (SELENE, Phase 1; fleet, Phase 3) and proven atmosphere subsystems (Chimborazo, Phase 2).

6. The Build Order

The lightning flashes before the thunder is heard: SELENE launches silently in vacuum before BRONTE shakes the air on Earth.

Phase	System	What gets built	What it proves / earns
1	**SELENE PoC** (Moon)	1–3 km surface track, 50–100 g, south-pole ridge	2.9 km @ 100 g clears lunar escape; exports water/regolith/metal to cislunar buyers from day one; grows to 10–12 km (30 g) then ~50 km (3 g)
2	**Chimborazo** (Earth)	14 km first build of the through-mountain line; evacuated tube, plasma window, aero-shell	Mach ~8.9 test capability at 6,160 m on the equator; DoD contracts fund operations; produces the real cost-per-km number
3	**Vacuum-world fleet**	SELENE-class kits on Phobos, Deimos, Ceres, the Belt	The core bet pays out: Mars' moons become freight terminals before ARES exists
4	**ARES** (Mars)	~180 km track, featherweight tube	Mars becomes a port; the freight network closes
5	**BRONTE full** (Earth)	Extend Chimborazo west to ~38 km (Mach 12.6); beyond that, the 1,000 km-class 3 g injector if economics prove out	The endgame — built only on validated cost data

SELENE is ~300× shorter than any full Earth injector and skips every atmosphere subsystem. Chimborazo Phase 1 is ~70× shorter and proves the hard atmosphere parts before any large commitment. They de-risk in parallel.

7. How the Network Moves Mass

Three launchers, one supply chain. Once they exist, moving mass around the inner system mostly costs what running the launchers costs.

- **Earth sends what only Earth can build** — machines, electronics, medicine. The eventual full injector runs 80 t/day (29,000 t/yr); the Chimborazo demonstrator's job is to earn that machine the right to exist. (Rev 1 wrongly pinned the 80 t/day cadence on the demonstrator itself.)
- **The Moon sends raw mass** — propellant feedstock, shielding regolith, structural metal, fired at 2.4 km/s toward cislunar catch points. At full cadence the cislunar propellant supply stops being constrained by what Earth can lift.
- **Mars closes the return trip** — ISRU methalox, samples, and exports ride ARES to Phobos/Deimos depots on electricity.
- **SEP tugs connect the nodes** — weeks per leg, nearly propellant-free; chemical engines do only final descents. The launchers do all the heavy lifting off every surface.
- **Each node funds the next.** Rockets deliver SELENE's first coils (the last rocket buy the lunar build needs); SELENE exports fund Chimborazo; Chimborazo revenue and data fund the fleet; the fleet builds ARES. Every airless body added makes the whole network richer.

8. The Skyhook: How Cost Falls Toward $5/kg

A rotating tether catches pods and flings them higher. No propellant. More payload per launch.

A long tether rotates in low orbit; its bottom tip sweeps backward, moving slower than orbital velocity. A pod that arrives at the tip gets grabbed; half a rotation later it is released up high, moving faster — to the Moon, GTO, or wherever the mission calls. With a 1.5 km/s tip speed, the launcher needs 6.3 km/s instead of 7.8: ~35% less energy, a slower exit (less heating), and the kick stage shrinks from ~20% of pod mass to 2–5% for trim — the freed 15–18% becomes payload. That shift is where $5/kg comes from.

- **Keeping it up:** electrodynamic reboost (current through the tether against Earth's field — demonstrated physics, no propellant) plus momentum returned by catching descending capsules.
- **The lunar extension:** a cislunar rotovator catches SELENE pellets and redirects them to LLO, L1, or Earth-transfer; an Earth-orbit tether receives. Cargo crosses cislunar space without burning its own propellant at all.
- **Provenance:** momentum-exchange tethers date to the 1970s; the Boeing/Tethers Unlimited HASTOL study (NASA NIAC, 1999–2001) detailed a 600 km tether with 3.5 km/s tip speed catching Mach-10 aircraft-launched payloads. Materials and manufacturing at scale, not new physics.

9. If the Full Earth Injector Gets Built

Not the plan — the vacuum fleet is the plan. But if the fleet works, this is what becomes possible.

A full Earth injector — ultimately the 1,000 km-class, 3 g, 8 km/s human-rated machine — is the most consequential infrastructure ever proposed. The key point: if SELENE works in vacuum and Chimborazo validates the atmosphere subsystems and the cost-per-kilometer, the full injector stops being a physics question and becomes a construction question.

What $5–20/kg to orbit changes

- **Satellites:** launch stops dominating cost — a 2,000 kg satellite rides for ~$20K at $10/kg vs the $83–97M a GPS III mission actually paid. Constellations replace on maintenance schedules.
- **Stations:** ISS-class logistics (~$3.1B/yr, ~$1.8B of it transport) collapses ~300×; thousand-person stations become economically normal.
- **Orbital manufacturing:** ZBLAN fiber, crystals, alloys — Flawless Photonics drew ~12 km of ZBLAN on ISS in 2024 (1,141 m in one day vs a 25 m prior record); the candid caveat is that published proof of reproducibly superior fiber is still pending. The queue is real either way, and it is waiting on freight prices.
- **Deep space:** with cheap Earth-to-orbit plus lunar propellant, Mars missions stop being nation-sized budget exercises; the Belt becomes commercially reachable.

> **The strategic picture.** The operator of an 80 t/day injector controls the access ramp to space — 29,000 t/yr can replenish a damaged constellation in hours. The host nation stops paying for space access and starts charging for it. The vacuum fleet pays for itself; the Earth injector is what the fleet eventually makes possible.

10. Zero New Physics

Every subsystem has flown, fired, or been demonstrated. The opportunity is assembly, not invention.

Building block	Status — verified June 12, 2026
EMALS (US Navy)	Operational on USS Gerald R. Ford since 2017 incl. combat deployment; 484 MJ flywheel storage, 45 s recharge. CVN-79 (Kennedy) completed builder's trials Feb 2026, delivery ~Mar 2027. 2023 AFOSR-filed report recommends evolving EMALS for lunar launch.
Superconducting maglev	603 km/h crewed record (JR Central L0, 2015) still stands; Shanghai Transrapid commercial since 2003; Chūō Shinkansen now expected NET ~2034 after the Shizuoka settlement (early 2026).
REBCO superconductors	Tc ≈ 92 K. All-REBCO records: 26.86 T (ASIPP, 2024) → **28.2 T** (CHMFL, 2025); all-superconducting record **35.1 T** (ASIPP, Sept 2025); 45.5 T hybrid (NHMFL, 2019). CFS demonstrated 20 T SPARC-class coils (2021); SPARC assembly underway (first TF coil placed Jan 2026), first-plasma targeted 2026, Q>1 targeted 2027; ARC (Virginia) early 2030s. Jc > 1,000 A/mm^2 at 30 T *at 4.2 K*. Launch LSM needs only 5–10 T at the gap.
MHD plasma window	Invented by A. Hershcovitch (Brookhaven), patented 1995; holds ~9 atm; **demonstrated** for non-vacuum e-beam welding at cm-scale apertures. Scaling to the Ø4 m bore (2.8× baseline power) is the program's R&D centerpiece — Phase 1 tests it at full bore.
X-59 shock shaping	First flight Oct 28, 2025; **first supersonic flight June 5, 2026** (Mach 1.1); Mach 1.4 acoustic points next; XVS camera-vision cockpit flight-proven.
High-g electronics	Gun-launched guidance routinely qualified to 15,000–20,000 g (Excalibur-class). SELENE's 100 g cargo mode is conservative by two orders of magnitude.
EM launch-assist market	Auriga Space: $12.2M cumulative incl. AFWERX D2P2; hypersonic-test-as-revenue model proven in market; outdoor track (Thor) planned 2026.
Orbital manufacturing demand	Flawless Photonics: ~11.9 km ZBLAN drawn on ISS (Feb–Mar 2024; 1,141 m/day vs 25 m prior record); Luxembourg production expansion; published performance characterization still pending — flagged honestly.
Lunar mass-driver physics	O'Neill, NASA Ames studies 1976–77; Mass Driver One demo 1977. SJSU sizing study (2023): 25.4 kg pellets, 2.4 km/s, 10 s cadence, 8.7 MW polar solar, ~1.0 MA per bucket coil.
Lunar intent at scale	SpaceX lunar pivot + mass-driver statements (Feb–Mar 2026). Founder remarks, not a program of record — but the direction of the largest launch company on Earth.
Fusion pull (He-3 context)	NIF ignition Dec 2022 (3.15 MJ); best shot now **8.6 MJ, target gain >4** (Apr 2025), ignition repeated 9+ times. Commercial He-3 demand is already contracted: Interlune–Bluefors offtake (Sept 2025) for up to 10,000 L/yr from 2028, prospecting mission 2027.

11. The Hard Parts

A document about a project this large that doesn't name the real problems is salesmanship, not engineering.

- **Plasma window scaling.** Centimeter demos → Ø4.0 m bore at 2.8× baseline power. First R&D dollar; if it doesn't scale, the muzzle needs a different answer and we must learn that at demonstrator scale, cheaply.
- **Cost per kilometer is unknown.** The $20–250/kg projection lives or dies on vacuum-rated superconducting track cost. Nobody has published a credible figure. Chimborazo Phase 1 produces the first real one.
- **The El Arenal piers.** 110–590 m A-frames in a 0.4 g seismic zone — beyond Millau's 343 m record. Partial trenching, rock-buttressed spans, or accepting a lower exit are the studied outs.
- **The reserve.** Chimborazo sits inside a protected fauna reserve under a rights-of-nature constitution. The legal path may be harder than the rock. Stated plainly because pretending otherwise would be fatal later.
- **SELENE's first bill is a rocket bill.** First coils, power, and radiators ride to the Moon at today's prices; the starter track must fit one or two heavy-lift flights.
- **Catching is harder than throwing.** Until the first cislunar catch happens, export revenue is theoretical. Wide-cone tugs first, tethers later.
- **Polar ice uncertainty.** Shackleton-specific reserves are unproven (bounded ≤5–10 wt% in surveyed regolith). The early business plan must close on regolith, oxygen, and metal even in the lean-ice case. Chang'e-7 data lands within a year.
- **The treaty question.** The Outer Space Treaty is silent on commercial mass drivers; a kinetic launcher is inherently dual-use. That ambiguity buys scrutiny on one side and AFWERX funding on the other. EMALS and GPS both walked this road.

12. What KERAUNOS Unlocks

Orbital manufacturing

- **ZBLAN fiber** — up to 100× lower theoretical loss than silica; ~12 km drawn on ISS in 2024; customers queue on freight prices. KERAUNOS is the freight price. (Performance proof still pending publication — the bet is stated, not oversold.)
- **Orbital bioprinting** — organs that collapse under their own weight in 1 g; daily return pods turn it from stunt to logistics business.
- **Debris salvage** — thousands of tonnes of aerospace alloy already in orbit becomes ore when tug-hours are cheap.

Geopolitical position

- **The Space Panama Canal** — the host nation stops paying for access and starts charging for it.
- **Presence as a supply chain** — 80 t/day secures orbital slots and lunar territory as a logistics fact, not a flag.
- **The deterrence dividend** — constellation replenishment in hours, not months.

Fusion economy

- **The He-3 supply chain.** Lunar regolith holds an estimated (model-based) ~1M tonnes of He-3. D-He3 fusion is **low-neutron** (D-D side reactions keep it from being strictly aneutronic) — far less activation, direct conversion potential. At today's ~$20M/kg (~$20B/tonne) market price and SELENE's ~$5/kg lunar launch cost, delivery economics are absurdly favorable; Interlune's contracted offtakes (Bluefors, 2028–2037) prove demand precedes the fusion era. SELENE isn't just a launcher — it's the fuel logistics for the energy industry after next.
- **The energy loop.** Trackside storage doubles as grid-scale battery between launch windows; the same plant powers debris-sweeping lasers over the corridor. Nothing idles.

Verification note: all external claims in this revision were re-checked against primary reporting and literature on June 12, 2026; all physics re-derived from first principles the same day. Annex A (BRONTE Chimborazo Design Map) is bound after Section 4. The SELENE Design Map (KER-SEL-TRK-002) is issued as a companion sheet.